

Progress Report: Sprint 5

CS.028 Optimizing Research Software Codes - 7 December 2025

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Executive Summary

- Additional function timing to determine the cause of the runtime issues.
- Completed the stakeholder presentation and report.
- Researched into creating a C/C++ to Python package to potentially create a package in C to parse the data faster.

Progress vs Plan

Planned Item	Status	Links	Notes/Blockers	Plan Change
Implement function testing	Partial	N/A	Still researching how to optimize the functions	Extend to next sprint
Documentation draft	Complete	GitHub	Architecture.md added for rnx2db.py	None
Program flowchart analysis	Partial	N/A	Currently analyzing the program to determine how many times each function is called	None
Add error handling	Partial	N/A	Error handling added for executable path	None
Prototype optimized functions	Not done	N/A	Explore and experiment possible program optimizations	None
Implement initial optimizations	Not done	N/A	Implement first program update	None
Add more sample data input	Complete	N/A	Project partner does not have additional data	None

Evidence of Working Software

CI/Build: [GitHub URL](#)

Risk and Quality

Risk	Owner	Status	Concrete Actions
Code complexity	Team	Active	Modulating code sections, adding inline comments
Setup inconsistencies	Team	Resolved	Documented environment dependencies
Limited testing time	Team	Active	Prioritize optimizing the slowest functions
Poor documentation of function tests	Team	Active	Log pre-optimization function summaries and their expected outputs
Hidden dependencies between functions	Team	Active	Use trace debugging tools to observe the inner workings of the original program
Unclear success criteria	Team	Resolved	Log all implemented optimizations with time test results

Bug count: zero current known errors, extremely poor runtime efficiency.

Next Goals

- Improve linting errors
- Increase inline documentation
- Analyze the problem functions further to determine any obvious runtime issues
- Review PRs and clean up the GitHub repository

Team Process Reflection

This sprint ended up going pretty well overall, even though it started a bit slowly. Once we got closer to the stakeholder presentation deadline, we were able to get the recording done. Working on that presentation actually helped us understand the system better, especially how the pieces of the database-loading pipeline fit together.

We still had some challenges, mainly with communication and timing. A lot of work happened very close to the deadline, which made it harder to coordinate and keep track of progress during the sprint. Going forward, we'll likely benefit from setting earlier checkpoints and breaking tasks into smaller steps so things don't pile up at the end.

Individual Contributions

Kathryn Butler

- Additional function timing to discover more details of the functions.
- Added evidence examples and artifacts to the Fall Term Retrospective.
- Created the stakeholder presentation slides and summary notes to deliver to the project partner.

Michael Mcallister

- Team Narrative Synthesis and Insights
- Theorized lookup table implementation for datasets
- Video editing, captioning, and time stamping stakeholder report

Joseph Schaab

- Created a note folder for documenting our understanding of the original code: [Link](#).
- Explored debugging options that will be needed for descriptive analysis of functions.
- Created stakeholder presentation slides and documented analysis information gathered this term.